Internship Report

Title / subject of the internship

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Internship carried out from 02/03/2026 to 04/09/2026

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Internship host company : IBISC (AROBAS) - Bâtiment IBGBI, 23 Bd de France 91034 Evry

Abstract

Write an abstract here.

Key words: financial models, machine learning...

Contents

1 Presentation of the company

2 Preliminary notions

2.1 Central Dogma

The Desoxyribose Nucléic Acid (DNA) is a fundamental moleculs in genomics. This molecule contains the genetics informations of each people. The flow of genetics informations

3 Introduction

4 Data and Dataset

4.1 PDB file structure

THE PDB file is a file to represent each atom from a molecule, especially for protein and RNA. Each line have a keywords to explicite

4.2 PDB database

5 Materials and Methods

5.1 Knowledge based score function

5.2 Objective function

5.3 Optimization algorithm

5.4 Complete Pipeline

5.5 Different Applications

6 Results

6.1 Interpolation results

6.2 Comparaison between original and predicted structures

6.3 Comparatives tables

6.4 Line chart of score function for folding trajectory

6.5 Internet platform for RNA folding

7 Discussion and Perspectives

8 Conclusion

Glossary

R

R

RASP Ribonucleic Acids Statistical Potential.

RNA RiboNucleic Acid.